SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I D. Dharshana (192524243) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Dharshana

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Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

ASSESSMENT TOOL - 3

1. Smart Home Automation system

A smart home automation system must intelligently control appliances and ensures user-comfort, energy savings, and security. Different intelligent agents can contribute in the following manner:

Sample Reflex Agent:

- Makes decisions based only on current sensor inputs.
- Uses IF - THEN rules.

Ex:-

- If motion detected → Turn ON lights
- If smoke detected → Trigger alarm.

Limitation: cannot remember previous events or learn user habit.

Mode - system

- Maintains an internals representation of the environment.
- Stores information such as:
  - → user's daily routhe
  - → previously occupied rooms.
  - → Historical temperature patterns.

Ex:-

If no motion is detected for 30 mins, the agent remembers the room is likely unoccupied and turns OFF the AC.

Goal - Based Agent:

- Operates with predefined objectives
- Goals include:
  - → maintanse indoor temperature b/w 22-25°C.
  - → keep the house secure.
  - → Reduce power consumption

Ex:-

Decideds whether the close blinds or turn on the air conditioner to achieve the desered temperature.

Utility - Based Agent

- evaluates multiple possible actions and select the one with the highest utility.
- considers:-
  - → comfort level
  - → electricity usage
  - → security level.
  - → user satisfaction

2] Robot Navigation in an unknown Maze

A robot must find an exit without prior knowledge of the maze. uniformed search algorithms can help in exploration.

Uniform cost search

- Expands the node with the lowest cummutive cost.
- Guarantees the least expensive path.

Advantages:-

- complete
- optimal for varyang path costs.
- suctable when movement costs differ.

Disadvantages:-

- High memory consumption.
- slower for large environments.

Complexity:

- Time complexity : High (depends on path cost)
- Space complexity : High.

Iterative ~~pdepending~~ deepending search

working:

- performance depth-limited searches repeatedly.
- Increases depth limit ~~skip~~ step by step.

Advantages:

- Requires very little memory.
- Finds shallow solutions efficiently

Disadvantages: -

- Re - explores nodes multiple times.
- less efficient for deep solutions.

Complexity:

- Time complexity : o(bd)
- Space complexity : o(bd)

Recommeded: Combination:

-> utility based agent + uniform cost search.
  -> find the least cost path
  -> optimizes battery usage.
  -> makes intelligent decisions.

Conclusion:

For an unknown maze, the combination of utility based agent and uniform cost search is the most effective because it ensure optimal path selection efficient resources utilization + intelligent decision making.

This approach improves the robot's ability to navigate complex environments successfully.